Assignment -7.5

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Batch -23

Task 1 (Mutable Default Argument – Function Bug)

Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

Bug: Mutable default argument

def add_item(item, items=[]):

items.append(item)

return items

print(add_item(1))

print(add_item(2))

Expected Output: Corrected function avoids shared list bug.

4 > 02 > 2026 > 7-1.py > ...

```python
#error Dont use mutable default argument
def add_item(item, items=None):
    if items is None:
        items = []
    items.append(item)
    return items
print(add_item(1,[10,20]))
print(add_item(2,[11.11, 22.22]))
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/
env/lab_05.py"
[10, 20, 1]
[11.11, 22.22, 2]
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 2 (Floating-Point Precision Error)

Task: Analyze given code where floating-point comparison fails. Use AI to correct with tolerance.

 # Bug: Floating point precision issue

def check_sum():

return (0.1 + 0.2) == 0.3

print(check_sum())

Expected Output: Corrected function

```
4 > 02 > 2026 > 7-2.py > ...
#error Floating-point precision
def check_sum():
    return abs((0.1 + 0.2) - 0.3) < 1e-10
print(check_sum())
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.venv/Scripts/python
env/lab_05.py"
True
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due tomissing base case. Use AI to fix.

 # Bug: No base case

def countdown(n):

print(n)

return countdown(n-1)

countdown(5)

Expected Output : Correct recursion with stopping condition.

```
4 > 02 > 2026 > 7-3.py > ...
#Error Recursion limit exceeded
def countdown(n):
    if n == 0:
        return
    print(n)
    countdown(n-1)
countdown(5)
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Ass
env/lab_05.py"
5
4
3
2
1
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error. Use AI to fix it.

Bug: Accessing non-existing key

def get_value():

data = {"a": 1, "b": 2}

return data["c"]

print(get_value())

Expected Output: Corrected with .get() or error handling.

CODE:

4 > 02 > 2026 > 7-4.py > ...

```python
#Error KeyError when accessing dictionary
# with a non-existent key
def get_value():
    data = {"a": 1, "b": 2}
    return data.get("c", "Key not found")
print(get_value())
```

OUTPUT:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.venv/Scr
env/lab_05.py"
Key not found
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 5 (Infinite Loop – Wrong Condition)

Task: Analyze given code where loop never ends. Use AI to detectand fix it.

Bug: Infinite loop

def loop_example():

i = 0

while i < 5:

print(i)

Expected Output: Corrected loop increments i.

4 > 02 > 2026 > 7-5.py > ...

```python
#Error in loop due to incorrect indentation
def loop_example():
    i = 0
    while i < 5:
        print(i)
        i += 1

loop_example()
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.ven
env/lab_05.py"
0
1
2
3
4
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to

fix it.

Bug: Wrong unpacking

a, b = (1, 2, 3)

Expected Output: Correct unpacking or using _ for extra values.

4 > 02 > 2026 > 7-6.py > ...

```python
# Error in unpacking values from a tuple
# using underscore for unused variable
a, b, _  = (1, 2, 3)
print(a, b)
```

OUTPUT:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/D
env/lab_05.py"
1 2
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 7 (Mixed Indentation – Tabs vs Spaces)

Task: Analyze given code where mixed indentation breaks execution. Use AI to fix it.

Bug: Mixed indentation

def func():

x = 5

y = 10

return x+y

Expected Output : Consistent indentation applied.

```
4 > 02 > 2026 > 7-7.py > ...
# Bug: Mixed indentation
def func():
    x = 5
    y = 10
    return x+y
print(func())
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.venv/Scrip
env/lab_05.py"
15
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 8 (Import Error – Wrong Module Usage)

Task: Analyze given code with incorrect import. Use AI to fix.

Bug: Wrong import

import maths

print(maths.sqrt(16))

Expected Output: Corrected to import math

```
4 > 02 > 2026 > 7-8.py
# Correct import
import math as m
print(m.sqrt(16))
```

OutPut:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.v
env/lab_05.py"
4.0
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 9 (Unreachable Code – Return Inside Loop)

Task: Analyze given code where a return inside a loop prevents full iteration. Use AI to fix it.

Bug: Early return inside loop

def total(numbers):

for n in numbers:

return n

print(total([1,2,3]))

Expected Output: Corrected code accumulates sum and returns

after loop.

```
4 > 02 > 2026 > 7-9.py > ...
# Task 9 (Logical Error - Incorrect Loop Logic)
def total(numbers):
    total_sum = 0
    for n in numbers:
        total_sum += n
    return total_sum
print(total([1,2,3]))
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Ass
env/lab_05.py"
6
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 10 (Name Error – Undefined Variable)

Task: Analyze given code where a variable is used before being defined. Let AI detect and fix the error.

Bug: Using undefined variable

def calculate_area():

return length * width

print(calculate_area())

Requirements:

• Run the code to observe the error.

• Ask AI to identify the missing variable definition.

• Fix the bug by defining length and width as parameters.

• Add 3 assert test cases for correctness.

Expected Output :

• Corrected code with parameters.

• AI explanation of the bug.

Successful execution of assertions.

4 > 02 > 2026 > 7-10.py > ...

```python
# Bug: Using undefined variable
def calculate_area():
    radius = 5
    area = 3.14 * radius * radius
    return area

print(calculate_area())
```

OutPut:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.venv/
env/lab_05.py"
78.5
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 11 (Type Error – Mixing Data Types Incorrectly)

Task: Analyze given code where integers and strings are added incorrectly. Let AI detect and fix the error.

Bug: Adding integer and string

def add_values():

return 5 + "10"

print(add_values())

Requirements:

• Run the code to observe the error.

• AI should explain why int + str is invalid.

• Fix the code by type conversion (e.g., int("10") or str(5)).

• Verify with 3 assert cases.

Expected Output #6:

• Corrected code with type handling.

• AI explanation of the fix. Successful test validation.

4 > 02 > 2026 > 7-11.py > ...

```python
def add_values():
    return 5 + int("10")

print(add_values())
# Test validation
assert add_values() == 15
assert 7 + int("3") == 10
assert int("20") + 5 == 25
print("All tests passed!")
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.ve
env/lab_05.py"
15
All tests passed!
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 12 (Type Error – String + List Concatenation)

Task: Analyze code where a string is incorrectly added to a list.

Bug: Adding string and list

def combine():

return "Numbers: " + [1, 2, 3]

print(combine())

Requirements:

• Run the code to observe the error.

• Explain why str + list is invalid.

• Fix using conversion (str([1,2,3]) or " ".join()).

• Verify with 3 assert cases.

Expected Output:

• Corrected code

• Explanation

• Successful test validation

```
4 > 02 > 2026 > 7-12.py > ...
def combine():
    return "Numbers: " + str([1, 2, 3])
print(combine())
assert combine() == "Numbers: [1, 2, 3]"
assert "List: " + str([4, 5]) == "List: [4, 5]"
assert "Values: " + str([]) == "Values: []"
```

OutPut:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi/.ven
env/lab_05.py"
Numbers: [1, 2, 3]
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 13 (Type Error – Multiplying String by Float)

Task: Detect and fix code where a string is multiplied by a float.

Bug: Multiplying string by float

def repeat_text():

return "Hello" * 2.5

print(repeat_text())

Requirements:

• Observe the error.

- Explain why float multiplication is invalid for strings.
- Fix by converting float to int.
- Add 3 assert test cases.

4 > 02 > 2026 > 7-13.py > ...

```python
def repeat_text():
    return "Hello" * int(2.5)

print(repeat_text())

assert repeat_text() == "HelloHello"
assert "Hi" * int(3.0) == "HiHiHi"
assert "A" * int(1.9) == "A"
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SA
env/lab_05.py"
HelloHello
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 14 (Type Error – Adding None to Integer)

Task: Analyze code where None is added to an integer.

Bug: Adding None and integer

def compute():

value = None

return value + 10

print(compute())

Requirements:

• Run and identify the error.

• Explain why NoneType cannot be added.

• Fix by assigning a default value.

• Validate using asserts.

CODE:

```
4 > 02 > 2026 > 7-14.py > ...
def compute():
    value = None
    if value is None:
        value = 0
    return value + 10

print(compute())

assert compute() == 10
assert (0 + 10) == 10
assert (5 + 10) == 15
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/D
env/lab_05.py"
10
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```

Task 15 (Type Error – Input Treated as String Instead of

Number)

Task: Fix code where user input is not converted properly.

Bug: Input remains string

def sum_two_numbers():

a = input("Enter first number: ")

b = input("Enter second number: ")

return a + b

print(sum_two_numbers())

Requirements:

- Explain why input is always string.
- Fix using int() conversion.
- Verify with assert test cases.

CODE:

4 > 02 > 2026 > 7-15.py > ...

```python
def sum_two_numbers(a, b):
    return int(a) + int(b)
print(sum_two_numbers("2", "3"))
assert sum_two_numbers("2", "3") == 5
assert sum_two_numbers("10", "5") == 15
assert sum_two_numbers("7", "8") == 15
```

OUTPUT:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv> & "C:/Users/SAICHARAN/Downloads/AI Assi
env/lab_05.py"
5
(.venv) PS C:\Users\SAICHARAN\Downloads\AI Assi\.venv>
```